

Haocheng Liu

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Experience

Two Sigma Investment LP

New York, NY

SOFTWARE ENGINEER

Dec. 2019 - Now

- **Streaming and enabling cloud capability for researching tick level market data**
 - Empowered the team to generate research market data as hive partitioning parquet in live with automated code generation and Acero execution planing for >100 feeds, adding up to hundres of terabytes daily.
 - Migrated research market data from a bespoke format to open source columnar data representation format Parquet, improving simulation speed use cases by 10x.
 - Enabled interative python researching on tick level market data via python bindings on C++ API as well as Apache execution engine, bridging Big table and Big Query through avro.
 - Built a generic C++ adapter for few hundreds market data readers and added python binding support.
- **Low latency martet data platform development and maintainence**
 - Lead a team of 4 engineers to expand company trading porfolio to Aisa, onboarding 6 direct feeds and backfill decades of research data.
 - Parsed real time, high-volume marketa data feeds from major stock exchanges all over the world, owner of future, FX, as well as asian equity feeds.
 - Redesigned fault tolerant tickerplant health reporting which addresses few sev0 incidents and roll it out across more than 100 feeds.
 - Add real time measurement to ISO order compliance framework via prometheus && Grafana and roll out to production without disruption to partners.
 - Modernalized the company C++ development toolchain by migrating bespoke build system to CMake, enabling static analysis build and onboarding new users.
- **Intraday market data platform modernalization**
 - Built a platform to support T+0 and T+1 use cases, visualizations, ad hoc queries at market data scale for all assert classes.
 - Collaborated with engineering partners to meet performance, auto failover, high availability goals of existing customers for criitical trading and modeling workloads.
 - Paired with cloud and solution vendors to migrate on-premises solutions to cloud, providing autoscaling of instances for scaling of new datasets and surges in queries.

Kitware Inc.

Clifton Park, NY

R&D ENGINEER

Jun. 2016 - Dec. 2019

- **Scientific multi-threading algorithms for Massively Threaded Architectures toolkit(VTK-m)**
 - Implemented a bunch of parallel scientific algorithms(probe, histogram, etc) and improved the shared memory support for CUDA and runtime selection for TBB and OpenMP
 - Simplified and optimized the testing infrastructure which saved runtime overheads by CRTP pattern and generic programming. It reduced the binary size by 80% and improved 40% of runtime performance
 - Refactored the code base with C++11 features which simplified meta programming, improving runtime overhead as well as adding a reliable random number generation and prepare for C++14 migration
- **Adaptable Input Output System Version 2(ADIOS2)**
 - Implemented a parallel data reader which could distribute and balance loads among many processes and visualize streaming data, in-situ data in large scale. It's capable of generating unstructured grid, uniform grid, polydata and volume data
 - Parallelized the engine tests(BP, HDF5, ADIOS1, etc) and binding tests(Python, Fortran), tested on hundreds of processor cores
 - Extended FindGoogleTesting CMake module so that it can launch MPI tests and provide a better CMake integration

Education

Carnegie Mellon University

Pittsburgh, PA

M.S. IN MECHANICAL ENGINEERING 3.7/4.0

Aug. 2014 - May. 2016

- Graduate research assistant, Visual Design and Engineering Lab
- Teaching assistant, Artificial Intelligent and Machine Learning

Shanghai Jiao Tong University

Shanghai, China

B.S. IN MARINE ENGINEERING 3.7/4.0

Sep. 2010 - June. 2014

- Outstanding graduate of Shanghai Jiao Tong University

Skills

DevOps	Linux, Multi threading, Meta programming, Docker, GDB, Vagrant, GCP, AWS
Framework	Qt, Google Testing, Visual Studio, Spack, Ansys APDL
Software	nlohmann_json, CUDA, OpenMP, Meshlab, TBB, OpenStreetMap, Pybind11
Open source	Apache Arrow, VTK, ParaView, SMTK, CMB, ADIOS2, Spack, CMake, Rasberry Pi
Programming	C/C++, CMake, Python, make, Git, Bash, LaTeX, Java, Fortran, groovy